1 2.6

Providing some motivation is good. Long time ago, with the aim of studying rational functions on compact Riemann surfaces S, the following space of functions was introduced

$$M(n_1p_1 + \cdots + n_kp_k) := \{\text{rational functions } f : S \to \mathbb{P}^1 \text{ such that } \operatorname{ord}_{p_i}(f) \geq -n_i\}.$$

Where $n_i \in \mathbb{Z}$, $p_i \in S$ and

$$\operatorname{ord}_p(f) = \begin{cases} a & f \text{ has a zero of order } a \text{ at } p \\ -a & f \text{ has a pole of order } a \text{ at } p \\ 0 & \text{Other} \end{cases}$$

Later, the notion of the divisor group was introduced,that is, $\operatorname{Div}(S) := \mathbb{Z}[S]$, whose elements are formal linear sums of points on S with integer coefficients. Then, to each rational function f, we associate a divisor denoted by $\operatorname{div}(f) := \sum_{s\in S} \operatorname{ord}_p(f) \cdot p$. This is a finite sum because S is compact, hence well-defined. A divisor D is called effective if each coefficient is non-negative, denoted by $D \geq 0$.

Then, let $D := n_1p_1 + \cdots + n_kp_k$ be a divisor, we can rewrite the above space of functions as follows

$$M(D) = \{\text{rational functions } f : S \to \mathbb{P}^1 \text{ such that } \operatorname{div}(f) + D \geq 0\}.$$

Riemann noticed that the dimension of this space is extremely important. He discovered an inequality that relates the dimension of the space of functions $h^0(D) := \dim_{\mathbb{C}} M(D)$, the numerical information of the divisor $\deg D := \sum n_i$ and the genus $g(S)$ of the compact Riemann surface S, as follows

$$h^0(D) \geq \deg D + 1 - g(S).$$

Later, Riemann's student Roch completed the inequality into an equality as follows

$$h^0(D) - h^0(K - D) = \deg D + 1 - g(S).$$

Where K denotes the canonical divisor on S, which is associated with the holomorphic 1-forms on S.

1.1 Cartier divisor

Let X be a variety over field k. Here, we develop the correspondence between Cartier divisors, the line bundle associated with a Cartier divisor, and the embedding of such bundles into the function field. We begin by considering codimension-one closed subschemes of X that are locally defined by a single function.

Consider an open cover $X = \bigcup_i U_i$, and pick $f_i \in \Gamma(U_i, \mathcal{O}_X) \subset K(X)$ satisfying $\frac{f_i}{f_j} \in \Gamma(U_i \cap U_j, \mathcal{O}_X)^\times$. According to our choice, the zero set of f_i

$$V(f_i) = \{x \in U_i | (f_i)_x \in \mathfrak{m}_x \subset \mathcal{O}_{X,x}\} \subset U_i$$

is a closed subscheme in U_i satisfy $V(f_i) \cap U_j = V(f_j) \cap U_i$. So we obtain a closed subscheme D on X. We denote $D = \{(U_i, f_i)\}_i$.

This collection of data induces a line bundle on X. For each U_i, we can define a subsheaf of the function field of U_i, that is, $\mathcal{O}_{U_i} \cdot \frac{1}{f_i} \subset \mathcal{K}_X|_{U_i}$, with transition functions

$$\varphi_{ij} : (\mathcal{O}_{U_i} \cdot \frac{1}{f_i})|_{U_i \cap U_j} \cong (\mathcal{O}_{U_j} \cdot \frac{1}{f_j})|_{U_i \cap U_j}$$
$$\frac{s_i}{f_i} \mapsto \frac{s_i}{f_i}.$$

Therefore these locally defined sheaves can be glued together to get a line bundle $\mathcal{O}_X(D)$ on X with isomorphism $\mathcal{O}_{\mathrm{X}}(D)|_{U_i} \cong \mathcal{O}_{U_i} \cdot \frac{1}{f_i}$. Each section $s \in \Gamma(V, \mathcal{O}_X(D))$ is of the form $s = \{(U_i \cap V, \frac{g_i}{f_i})\}$ such that $\frac{g_i}{f_i} = \frac{g_j}{f_j}$ on $U_i \cap U_j \cap V$. Similarly, by gluing the local sheaf $\mathcal{O}_{U_i} \cdot f_i$ together, we obtain an idea sheaf $\mathcal{I}_D \subset \mathcal{O}_X$. We can see that $\mathcal{I}_D$ is also a line bundle such that $\mathcal{O}_X(D) \otimes \mathcal{I}_D \cong \mathcal{O}_{\mathrm{X}}$ and $D = V(\mathcal{I}_D)$.

Let's descrive the embedding of $\mathcal{O}_X(D)$ into the sheaf of rational functions $\mathcal{K}_X$. Note that we have an injection

$$\mathcal{O}_X \hookrightarrow \mathcal{K}_X.$$

Tensoring $\mathcal{O}_X(D)$, we have

$$\mathcal{O}_X(D) \hookrightarrow \mathcal{O}_X(D) \otimes \mathcal{K}_X.$$

If $D = \{(U_i, f_i)\}$, then we obtain an isomorphism as following

$$\mathcal{O}_X(D) \otimes \mathcal{K}_X \cong \mathcal{K}_X$$
$$\{(U_i \cap V, \frac{g_i}{f_i} \otimes a)\} \mapsto \{(U_i \cap V, a \cdot \frac{g_i}{f_i})\}.$$

Note that these locally defined functions $\{(U_i \cap V, a \cdot \frac{g_i}{f_i})\}$ patch together to define a rational function $a \cdot \frac{g}{f} \in \Gamma(V, \mathcal{K}_X) = K(X)$ such that $(a \cdot \frac{g}{f})|_{U_i} = a \cdot \frac{g_i}{f_i}$. Combined with this isomorphism, we have

$$\mathcal{O}_X(D) \hookrightarrow \mathcal{K}_X.$$

Therefore, the sections of $\mathcal{O}_X(D)$ can be equivalently desctibed as follows

$$\begin{aligned}\Gamma(V, \mathcal{O}_X(D)) &= \{f \in K(X) \text{ such that } f|_{U_i} \cdot f_i \in \mathcal{O}_{U_i}\} \\ &= \{f \in K(X) \text{ such that } f|_V \cdot \Gamma(V, \mathcal{I}_D) \subset \Gamma(V, \mathcal{O}_X)\} \\ &= \{f \in K(X) \text{ such that } \operatorname{div}(f)|_V + D|_V \geq 0\}.\end{aligned}$$

Where $\operatorname{div}(f) = \{(X, f)\}$ and $\operatorname{div}(f)|_V + D|_V = \{(U_i, f|_{U_i} \cdot f_i)\} \geq 0$ means that $f|_{U_i} \cdot f_i \in \mathcal{O}_{U_i}$. So we can see that the sections of $\mathcal{O}_X(D)$ are the rational functions that has no poles worse that D.

Example 1.1.1. *Setting $X = \mathbb{P}^1_k$, $U_0 = D_+(x_0) = \operatorname{Spec} k[\frac{x_1}{x_0}]$, $U_1 = D_+(x_1) = \operatorname{Spec} k[\frac{x_0}{x_1}]$, $f_0 = \frac{x_1}{x_0}$, $f_1 = 1$. Then we have a Cartier divisor $D = \{(U_0, f_0), (U_1, f_1)\}$. It determines a closed subscheme of codimension one $[1 : 0] \in \mathbb{P}^1$ and it provides a line bundle $\mathcal{L}$ such that $\mathcal{L}|_{U_i} = \mathcal{O}_{U_i} \cdot \frac{1}{f_i}$. Let $s \in \Gamma(\mathbb{P}^1, \mathcal{L})$ be a section. It is represented by*

$$s = \{\big(U_0, \frac{a_0 + a_1\frac{x_1}{x_0} + \cdots}{\frac{x_1}{x_0}}\big), \big(U_0, \frac{b_0 + b_1\frac{x_0}{x_1} + \cdots}{1}\big)\}.$$

Since $\frac{a_0 + a_1\frac{x_1}{x_0} + \cdots}{\frac{x_1}{x_0}} = \frac{b_0 + b_1\frac{x_0}{x_1} + \cdots}{1}$, we have $a_0 = b_1$, $a_1 = b_0$, $a_2 = a_3 = \cdots = b_2 = b_3 = \cdots = 0$. So we can see that

$$\Gamma(\mathbb{P}^1, \mathcal{L}) = \{a_1 + a_0\frac{x_0}{x_1} \in K(\frac{x_1}{x_0})\} = \{a_0x_0 + a_1x_1 \in k[x_0, x_1]\}.$$

If we pick $f_0 = 1$, $f_1 = \frac{x_0}{x_1}$ and let $\mathcal{L}'$ be the corresponding line bundle, it can be shown in the same manner as above that

$$\Gamma(\mathbb{P}^1, \mathcal{L}') = \{a_1\frac{x_1}{x_0} + a_0 \in K(\frac{x_1}{x_0})\} = \{a_0x_0 + a_1x_1 \in k[x_0, x_1]\}.$$

A difference arises here. While it can be shown that $\mathcal{L} \cong \mathcal{L}'$, the images of their respective embeddings into the function field are two distinct sub-sheaves.

Let $\mathcal{L}$ be an arbitrary line bundle on X then $\mathcal{L} \otimes \mathcal{K}_X \cong \mathcal{K}_X$, therefore $\mathcal{L}$ admits an embedding into $\mathcal{K}_X$ given by $\mathcal{O}_X \subset \mathcal{K}_X$. Suppose there exists another injection

$$\mathcal{L} \hookrightarrow \mathcal{K}_X.$$

Let $\{U_i\}$ be an open cover such that $\mathcal{L}|_{U_i} \cong \mathcal{O}_{U_i}$, let $\frac{1}{f_i} \in K(X)$ denote the image of the local generation of this injection $\mathcal{L}|_{U_i} \to \mathcal{K}|_{U_i}$. It follows that the embedding factors through $\mathcal{L} \to \mathcal{O}_X(D) \hookrightarrow \mathcal{K}_X$, where the first arrow is injective and $D = \{(U_i, f_i)\}$. Hence, we obtain an isomorphism

$$\mathcal{L} \cong \mathcal{O}_X(D).$$

If there are two embeddings $i_1, i_2 : \mathcal{L} \to \mathcal{K}_X$ with corresponding Cartier divisors D_1, D_2 such that $\mathcal{L} \cong \mathcal{O}_X(D_1) \cong \mathcal{O}_X(D_2)$. What happens between D_1 and D_2?

Let $\{U_i\}$ be a common open cover such that $\mathcal{O}_X(D_1)|_{U_i} \cong \mathcal{O}_X(D_2)|_{U_i} \cong \mathcal{O}_{U_i}$. Note that there exist $u_{ij} \in \mathcal{O}_X(U_i \cap U_j)^\times$ making the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_{U_i \cap U_j} & \xrightarrow{\times u_{ij}} & \mathcal{O}_{U_i \cap U_j} \\ {\scriptstyle \phi_i|_{U_j}}\uparrow & & \uparrow{\scriptstyle \phi_j|_{U_j}} \\ \mathcal{L}_{U_i \cap U_j} & \xrightarrow{=} & \mathcal{L}_{U_i \cap U_j}. \end{array}$$

Where $\{\phi_i\}$ is a collection of local trivializations of $\mathcal{L}$. Then we get

$$\frac{1}{f_j} = u_{ij} \cdot \frac{1}{f_i},$$
$$\frac{1}{g_j} = u_{ij} \cdot \frac{1}{g_i}.$$

That is

$$\frac{f_i}{g_i} = \frac{f_j}{g_j}.$$

Let $h \in K(X)$ be a rational function such that $h|_{U_i} = \frac{f_i}{g_i}$, we obtain that

$$D_1 - D_2 = \operatorname{div}(h).$$

1.2 Weil divisor

Let X be a variety over field k. Recall that an effective Cartier divisor of X can be viewed as a closed subscheme of X such that it is locally cut out by a single function. But not every codimensional-one closed integral subscheme is of this form. In fact, we have

Proposition 1.2.1. *Let X be the variety as above. Then the following are equivalent.*

(1) X is locally factorial.

(2) Every closed integral subscheme of X is locally defined by a single function.

This proposition is just the geometric version of the following lemma

Lemma 1.2.2. *A a noetherian integral domain. Then A is factorial if and only if every prime of height 1 is a principal ideal.*

Now, let us define the associated sheaf of a Weil divisor.

Definition 1.2.3. *Let X be a normal variety, $K(X)$ the function field. Given a Weil divisor D, define $\mathcal{O}_X(D)$ to be the sheaf such that*

$$\Gamma(V, \mathcal{O}_X(D)) = \{f \in K(X) | \operatorname{div}(f)|_V + D|_V \geq 0\} \cup \{0\}.$$

By "$\geq$", I mean that $\operatorname{div}(f)|_V + D|_V$ is effective on V.

Note that $\mathcal{O}_X(D)$ is in fact a coherent sheaf, though I omit the proof here.

Let U be the regular locus of the normal variety X, let $j : U \to X$ denote the open immersion. Then $\mathcal{O}_X(D)|_U$ restricts to a line bundle over U. Moreover we have

$$\begin{aligned} j_*(\mathcal{O}_X(D)|_U) &\cong j_*(\mathcal{O}_X(D)|_U^{\vee\vee}) \\ &\cong j_*\mathcal{H}om_{\mathcal{O}_U}(\mathcal{O}_X(D)|_U^{\vee}, \mathcal{O}_U) \\ &\cong \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X(D)^{\vee}, j_*\mathcal{O}_U) \\ &\cong \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X(D)^{\vee}, \mathcal{O}_X) \\ &\cong \mathcal{O}_X(D)^{\vee\vee}. \end{aligned}$$

And this isomorphism makes the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_X(D) & \longrightarrow & \mathcal{O}_X(D)^{\vee\vee} \\ & \searrow & \downarrow \cong \\ & & j_*\mathcal{O}_X(D)|_U. \end{array}$$

So we can see that $\mathcal{O}_X(D)$ satisfies the Hartogs' extension property (that is, the canonical map $\mathcal{O}_X(D) \to j_*\mathcal{O}_X(D)|_U$ is an isomorphism) if and only if it is a reflexive sheaf of rank one.

Definition 1.2.4. *Let X be a vairety over k. A coherent sheaf $\mathcal{F}$ is called reflexive, if the canonical map $\mathcal{F} \to \mathcal{F}^{\vee\vee}$ is an isomorphism. By a reflexive sheaf of rank n, I mean a reflexive sheaf whose stalk at the generic point is a vector space of dimension n.*

Proposition 1.2.5. *Let X be a normal variety over k, U the regular locus, D a Weil divisor on X and $\mathcal{O}_X(D)$ the associated coherent sheaf of D. Then $\mathcal{O}_X(D) \cong j_*\mathcal{O}_X(D)|_U$.*

Proof. It suffices to prove that the restriction map $\rho : \Gamma(V, \mathcal{O}_{\mathrm{X}}(D)) \to \Gamma(V \cap U, \mathcal{O}_{\mathrm{X}}(D))$ is an isomorphism for each open subset $V \subset X$. Let $f \in K(X)$ be a rational function. If f restricts to 0 on $V \cap U$, then $f = 0$ since X is integral, this implies the injectivity of ρ. If f satisfies $\mathrm{div}(f)|_{V\cap U} + D|_{V\cap U} \geq 0$, then it follows that $\mathrm{div}(f)|_V + D|_V \geq 0$ since $\mathrm{Codim}_V(V \setminus (V \cap U)) \geq 2$, this implies the surjectivity of ρ. □

So we can see that $\mathcal{O}_{\mathrm{X}}(D)$ is a reflexive sheaf of rank one according to the above discussion. Since the sections of a reflexive sheaf $\mathcal{F}$ come from its "locally free part" $\mathcal{F}|_U$, we can embed a reflexive sheaf of rank one as a subsheaf of the constant sheaf $\mathcal{K}_X$ as follows.

Proposition 1.2.6. *Any reflexive rank one sheaf $\mathcal{F}$ on a normal variety X embeds as a subsheaf of $\mathcal{K}_X$.*

Proof. Let U be the regular locus and $j : U \to X$ the open immersion. Then we have

$$\mathcal{F} \xrightarrow{\cong} j_*\mathcal{F}|_U \longrightarrow j_*\mathcal{K}_X|_U \xrightarrow{\cong} \mathcal{K}_X$$

The middle map is injective. □

Proposition 1.2.7. *Let $\mathcal{F}$ be a reflexive rank one sheaf on a normal variety X. Then there exist a Weil divisor D such that $\mathcal{F} \cong \mathcal{O}_X(D)$.*

Proof. Note that there exist a Cartier divisor D_U on U such that $\mathcal{F}|_U \cong \mathcal{O}_U(D_U)$. Then there exist a Weil divisor D such that $D|_U = D_U$. Since we have

$$\begin{aligned}\Gamma(V, \mathcal{O}_U(D_U)) &= \{f \in K(X) | \, \mathrm{div}(f)|_V + D_U|_V \geq 0\} \cup \{0\} \\ &= \Gamma(V, \mathcal{O}_X(D)|_U),\end{aligned}$$

it follows that $\mathcal{O}_U(D_U) \cong \mathcal{O}_X(D)|_U$. We thus obtain

$$\mathcal{F} \cong j_*(\mathcal{F}|_U) \cong j_*(\mathcal{O}_U(D_U)) \cong j_*(\mathcal{O}_X(D)|_U) \cong \mathcal{O}_X(D),$$

as desired. □

1.3 Example

Example 1.3.1. *Let X be a regular in codimension-one k-variety and let $\pi : X \times \mathbb{A}^1 \to X$ be the projection. Then $X \times \mathbb{A}^1$ is also a regular in codimension-one variety and $\mathrm{Cl}(X \times \mathbb{A}^1) \cong \mathrm{Cl}(X)$.*

First, we must show that there are only two types of codimension-one closed subschemes on $X \times \mathbb{A}^1$, that is, 'vertical' and 'horizontal' divisors. By taking an affine open cover $\{U_i\}$ of $X \times \mathbb{A}^1$, the problem reduces to determining the height one prime ideal of $R[t]$. Let K be the fraction field of R.

Let $\mathfrak{B} \in \operatorname{Spec} R[t]$ be a height one prime ideal. Since $\mathrm{ht}_R(\mathfrak{B} \cap R) \leq \mathrm{ht}_{R[t]}(\mathfrak{B})$, there are only two cases to consider.

If $\mathrm{ht}_R(\mathfrak{B} \cap R) = 1$, let $\mathfrak{p} = \mathfrak{B} \cap R$, we get a chain of prime ideal

$$(0) \subset \mathfrak{p}R[t] \subset \mathfrak{B}.$$

It follows that $\mathfrak{p}R[t] = \mathfrak{B}$, a vertical divisor. Note that

$$R[t]_{\mathfrak{p}R[t]} = (R_{\mathfrak{p}}[t])_{\mathfrak{p}R_{\mathfrak{p}}[t]},$$

where $R_{\mathfrak{p}}$ is a DVR and hence $\mathfrak{p}R_{\mathfrak{p}}[t]$ is a principal ideal, which implies that $(R_{\mathfrak{p}}[t])_{\mathfrak{p}R_{\mathfrak{p}}[t]}$ is also a DVR.

If $\mathrm{ht}_R(\mathfrak{B} \cap R) = 0$, then $\mathfrak{B} \cap R = (0)$. Let $S = R \setminus \{0\}$, then $S^{-1}R[t] = K[t]$ and $\mathfrak{B}K[t] = (f(t))$, where $f \in K[t]$. So we have $\mathfrak{B} = R[t] \cap f(t)K[t]$, a horizontal divisor. Note that we have

$$R \setminus \{0\} \subset R \setminus \mathfrak{B}.$$

So we get

$$R[t]_{\mathfrak{B}} = K[t]_{(f)},$$

a DVR.

We next prove that any horizontal divisor is linearly equivalent to a sum of vertical divisors. Let $a \in R$ be a regular element such that $a \cdot f \in R[t]$, then it follows that $\mathfrak{B}R_a[t] = (f(t))$. Since $V(a) \subset \operatorname{Spec} R[t]$ is a codimension one closed subscheme, we have the following sequence

$$\bigoplus_{F_i} \mathbb{Z}[F_i] \to \mathrm{Cl}(R[t]) \to \mathrm{Cl}(R_a[t]) \to 0.$$

Where the F_i are irreducible components of $V(a)$. Let D be the divisor on $\operatorname{Spec} R[t]$ corresponding to the ideal $\mathfrak{B}$. Since the image of D in $\operatorname{Cl}(R_a[t])$ is 0, we have

$$D \sim \sum_i n_i[F_i],$$

as expected.

Note that we have a map $\pi^* : \operatorname{Cl}(X) \to \operatorname{Cl}(X \times \mathbb{A}^1)$ given by

$$D = \sum_i n_i[Y_i] \mapsto \sum_i n_i[\pi^{-1}(Y_i)].$$

The discussion above implies that π^* is surjective. If $\pi^* D = \sum_i n_i[\pi^{-1}(Y_i)] = \operatorname{div}(f)$ for some $f \in K(t)$. Let $U = X \setminus (\bigcup_i Y_i)$ we see that $\operatorname{div}(f)|_{U \times \mathbb{A}^1} = 0$ on $U \times \mathbb{A}^1$. It follows that f is an invertible element in $K[t]$. So $f \in K^\times$ and $D = \operatorname{div}(f)$. So π^* is an injection.

Example 1.3.2. $\operatorname{Cl}(X \times \mathbb{P}^1) \cong \operatorname{Cl}(X) \times \mathbb{Z}$.

We have the following sequence

$$\mathbb{Z}[X_\infty] \to \operatorname{Cl}(X \times \mathbb{P}^1) \to \operatorname{Cl}(X \times \mathbb{A}^1) \to 0.$$

Where $X_\infty := X \times \{\infty\}$. First, we show that $n[X_\infty] = \operatorname{div}(f)$ implies $n = 0$. Since $\operatorname{div}(f)|_{X \times \mathbb{A}^1} = 0$ implies that $f \in K(X)[t]^\times = K(X)^\times$, we have $\operatorname{div}(f) = \sum_i n_i[\pi^{-1}(Y_i)] \in Z^1(X \times \mathbb{A}^1)$. Then we have $n = n_i = 0$.

Let $\pi : X \times \mathbb{P}^1 \to X$ be the projection. The map

$$\operatorname{Cl}(X) \cong \operatorname{Cl}(X \times \mathbb{A}^1) \to \operatorname{Cl}(X \times \mathbb{P}^1)$$

splits the sequence.